Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

STAFF ANSWERS · Supported

Arrival

90 s is faster.

Orange-brown.

1 ÷ 100 = 0.0100 s^{-1} (0.01 s^{-1} is also correct).

EVALUATE.

Exit

1/120 = 0.00833 s^{-1}; pH 7 is faster in the first series (90 s).

The first statement is supported. Only pH 4–8 in these conditions were tested; the repeat also shows variation.

Supported independent task

1. pH 6: 1/120 = 0.00833 s^{-1}. pH 7: 1/90 = 0.0111 s^{-1}.

2. Plot (6, 0.00833) and (7, 0.0111), pH on x-axis, relative rate/s^{-1} on y-axis. Accept equivalent pupil-directed placement.

3. pH 7 is fastest in the first series, because 0.0111 s^{-1} is the highest measured relative rate (or 90 s the shortest time).

4. The exact endpoint between samples is unknown. A shorter fixed interval could narrow the timing window if authorised and feasible; it does not remove every uncertainty.

5. Starch breaking into smaller sugars: digestion. Particles moving down a concentration gradient: diffusion.

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STAFF ANSWERS · Independent

Arrival

The same measured change takes less time; 1/90 is larger than 1/180.

90 s, while acknowledging the endpoint was reached after 60 s and by 90 s.

Relative rate = 1/120 = 0.00833 s^{-1}.

Describe states how values change; explain uses biology, such as pH affecting the active site, to account for the pattern.

Exit

pH 6: 0.00833 s^{-1}; pH 7: 0.0111 s^{-1}. pH 7 is faster in the first series under these matched conditions.

Repeat matched trials to assess variation, or use shorter authorised sampling intervals to narrow the 30 s endpoint window.

Independent independent task

1. Relative rate = 1/time. pH 6: 1/120 = 0.00833 s^{-1}. pH 7: 1/90 = 0.0111 s^{-1}. The shorter time produces the larger reciprocal rate.

2. x-axis pH (no unit); y-axis relative rate/s^{-1}. Points 4:0.00417, 5:0.00556, 6:0.00833, 7:0.0111, 8:0.00667. Rate rises from pH 4 to 7, then falls at pH 8.

3. pH 7 has the highest first-series rate, 0.0111 s^{-1} versus e.g. pH 4 at 0.00417 s^{-1}. pH can alter the active-site environment/shape and affect substrate binding and catalysis. The graph does not prove permanent denaturation or an exact universal optimum.

4. Strength: matched starting amounts/temperature and a common endpoint support comparison. Limitation: 30 s intervals or subjective colour judgement. Improvement: shorter fixed intervals if authorised, or consistent agreed colour judgement; link it to the limitation. The pH 7 repeat gives 120 s, so variation weakens confidence from one series.

5. Digestion is chemical breakdown catalysed by enzymes; amylase breaks starch into smaller sugars. Diffusion is net particle movement from higher to lower concentration. Moving particles does not itself change starch molecules chemically.

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STAFF ANSWERS · Aspire

Arrival

Its relative rate halves.

It summarises time to an observed endpoint, with sampling and colour limits. It does not continuously measure the amount changed per instant.

1/120 = 0.00833 s^{-1}; 1000/120 = 8.33 on that stated scaled convention. They must not be mixed on one unlabelled axis.

For example: one trial per pH limits repeatability evidence, so repeat matched trials before treating the peak as secure.

Exit

Repeat rate = 1/120 = 0.00833 s^{-1}, matching first-series pH 6. It shows variation and weakens an exact-optimum claim.

Shorter authorised intervals narrow the timing window, but this graph measures neither active-site structure nor recovery and cannot prove permanent denaturation.

Aspire independent task

1. pH 6 = 1/120 = 0.00833 s^{-1}. First pH 7 = 1/90 = 0.0111 s^{-1}. Separate repeat pH 7 = 1/120 = 0.00833 s^{-1}; it equals the first pH 6 rate. Keep both pH 7 observations visible.

2. First-series graph uses the five supplied/calculated coordinates. Repeat belongs at (7, 0.00833), as the same IV condition measured again, with its own symbol/key. Do not silently replace or average raw points.

3. The claim is too broad. pH 7 is highest only in the first series among pH 4–8 tested. A 120 s repeat shows variation and matches the first pH 6 time. Untested values and other amylase sources/conditions remain unknown. pH effects on active sites are a biological explanation, not direct molecular observations from this graph.

4. Matched starting amounts/temperature and common endpoint strengthen validity. A 30 s interval limits timing precision; a shorter fixed interval could narrow that gap if feasible/authorised. It does not remove subjective colour judgement, variable starting conditions or all repeat variation.

5. Digestion chemically breaks large molecules down; diffusion moves particles down a concentration gradient. They are different processes. The graph measures an endpoint in starch breakdown, not membrane transport, concentration gradients across a gut wall or transport proteins.

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STAFF ANSWERS · shared tasks and mark scheme

Shared activities

Starter · reverse the time: 90 s is faster than 180 s for matched amounts and the same endpoint. 1/90 ≈ 0.0111 s^{-1}; 1/180 ≈ 0.00556 s^{-1}. Doubling the time halves the relative rate. Worked rate and graph point: Relative rate = 1/time = 1/120 = 0.00833 s^{-1}. Plot pH 6 at 0.00833 s^{-1}; pH has no unit. Build an evidence chain: Calculation checks: a reciprocal value; shorter time gives larger rate. Pattern/conclusion: pH 7 has the highest first-series rate among tested values; rate rises to pH 7 then falls at pH 8. Evaluation: colour judgement, shorter sampling intervals, and transparent treatment of repeats/anomalies. Graph and claim repair: First-series rates are 0.00417, 0.00556, 0.00833, 0.0111 and 0.00667 s^{-1} for pH 4–8. The separate pH 7 repeat is 0.00833 s^{-1}, plotted with a distinct symbol at x = 7. These values do not prove an exact universal optimum or permanent denaturation. Topic 1 process labels: Starch chemically breaking into smaller sugars is enzyme digestion. Net movement of particles from higher to lower concentration is diffusion. Movement does not itself chemically break starch into sugars.

Independent

1. Calculate relative rate for pH 6 and pH 7, showing equation, substitution and units. Check that shorter time gives a larger value. [4 marks]

Use relative rate = 1/time (1 mark)

1/120 = 0.00833 s^{-1} (1 mark)

1/90 = 0.0111 s^{-1} (1 mark)

The shorter time gives the larger relative rate (1 mark)

Independent

2. Complete the five-point graph using the two calculated rates and three supplied points. Label both axes and state the rise-and-fall pattern. [4 marks]

Plot pH 6 at 0.00833 s^{-1} (1 mark)

Plot pH 7 at 0.0111 s^{-1} (1 mark)

Label pH and relative rate / s^{-1} (1 mark)

The rate rises to pH 7 then falls at pH 8 (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Independent

3. Identify the highest tested activity in the first series using one numerical comparison. Explain how pH can affect the enzyme's active site. [3 marks]

pH 7 has the highest first-series rate, 0.0111 s^{-1} (1 mark)

Support with another value, such as pH 6 at 0.00833 s^{-1} (1 mark)

pH can change active-site shape or binding and so enzyme activity (1 mark)

Independent

4. Give one strength, one limitation and one linked improvement for the endpoint method. Use the separate pH 7 repeat of 120 s to qualify your confidence. [4 marks]

A consistent observable endpoint or matched conditions is a strength (1 mark)

Sampling gaps or subjective colour judgement is a limitation (1 mark)

A shorter authorised interval or agreed colour standard targets that limitation (1 mark)

The pH 7 repeat of 120 s differs from 90 s and limits repeatability confidence (1 mark)

Independent

5. Distinguish enzyme digestion from diffusion in the supplied two-step diagram. Explain why movement from high to low concentration does not itself break starch into sugars. [3 marks]

Digestion is chemical breakdown of starch into smaller molecules (1 mark)

Diffusion is net movement down a concentration gradient (1 mark)

Movement itself does not break the starch molecule (1 mark)

Aspire

1. Calculate pH 6 and pH 7 relative rates. Compare the first pH 7 rate with its separate 120 s repeat, retaining both results. [4 marks]

1/120 = 0.00833 s^{-1} at pH 6 (1 mark)

1/90 = 0.0111 s^{-1} at pH 7 (1 mark)

The separate pH 7 repeat is 0.00833 s^{-1} (1 mark)

Retain both pH 7 trials (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Aspire

2. Complete all five points on the first-series graph. Add the separate repeat at the same pH 7 x-position. Use a different symbol for it. Explain why it is not pH 9. Explain why it must not just replace a point without saying so. [4 marks]

Plot the five first-series points with correct axes (1 mark)

Plot the repeat at pH 7 and 0.00833 s^{-1} (1 mark)

Use a distinct symbol/key (1 mark)

The repeat has the same pH and is not a replacement observation (1 mark)

Aspire

3. Evaluate “pH 7 is the exact optimum for every amylase”. Use the tested range, the repeat and the difference between data and a molecular explanation. [4 marks]

pH 7 is highest only among the values tested (1 mark)

An exact optimum between tested values was not measured (1 mark)

The repeat shows variation (1 mark)

Molecular mechanism is an explanation, not directly observed in the timing data (1 mark)

Aspire

4. Give a strength–limitation–improvement chain. Explain which uncertainty a shorter authorised sampling interval reduces and which it leaves. [4 marks]

Identify a relevant strength (1 mark)

Name a sampling-gap limitation (1 mark)

Link a shorter authorised interval to a narrower timing window (1 mark)

It does not remove subjective colour judgement or all systematic error (1 mark)

Aspire

5. Explain why “diffusion digests starch” mixes up two processes. Use the definitions given. State why an enzyme-rate graph on its own cannot show how sugars cross a gut membrane. [4 marks]

Digestion changes starch molecules into smaller sugars (1 mark)

Diffusion moves particles down a concentration gradient (1 mark)

The graph measured starch disappearance rather than membrane transport (1 mark)

It cannot establish how sugars cross the gut membrane (1 mark)

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STAFF ANSWERS · shared tasks and mark scheme · continued

Aspire

Accept scientifically equivalent wording. Award each distinct point once. The mark tariff is for the question, not a participation score. Keep the answer scheme staff-only during independent work.

Choice

Use the matching independent-task answers; preserve the pupil's own scientific sentence, label or reason. Staff mark I, S or C beside this task.